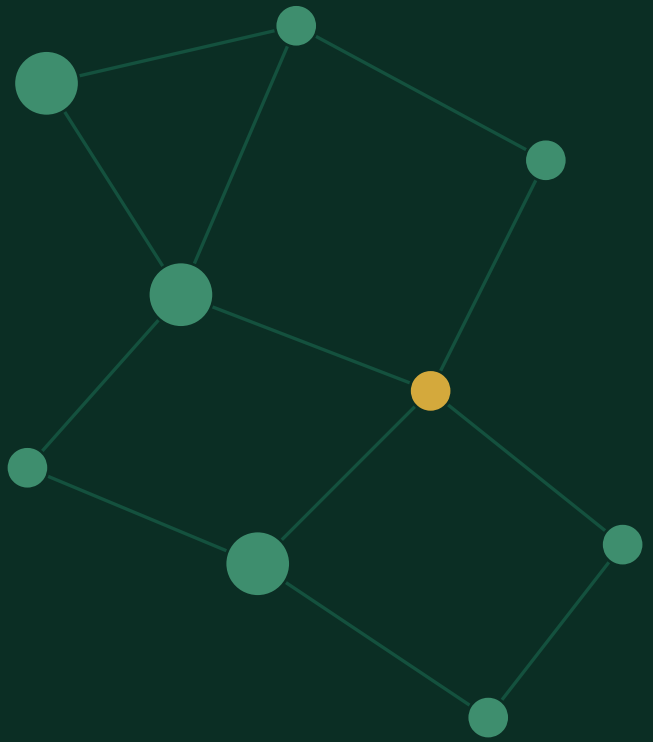
PROJECT DEFENSE

AraCare Health OS

A hospital operating system built for how Nigerian facilities actually run.

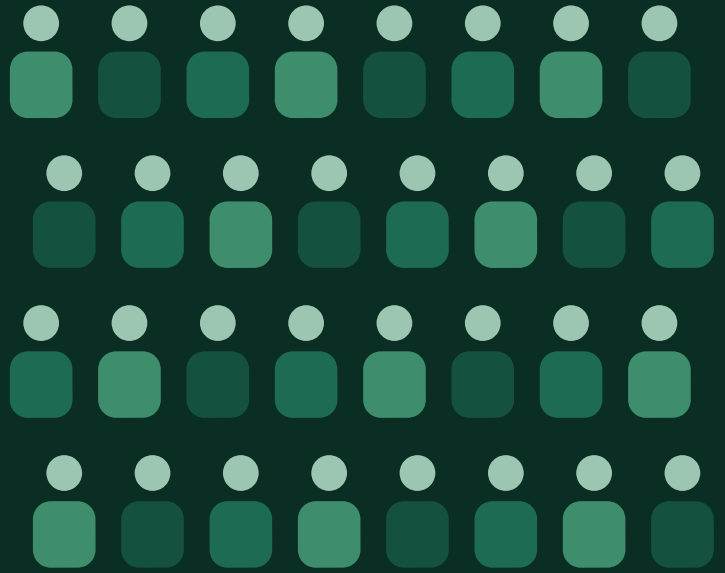


Joseph Cosmos Aniebiet · 22/SC/CO/1301 · Computer Science

Before sunrise, the line has formed

Her child has a fever. She joins the line early, because she knows what the day will look like.

There is no ticket, no order and no estimate. Place in line is decided by who pushes in first, and every patient waits without knowing for how long.

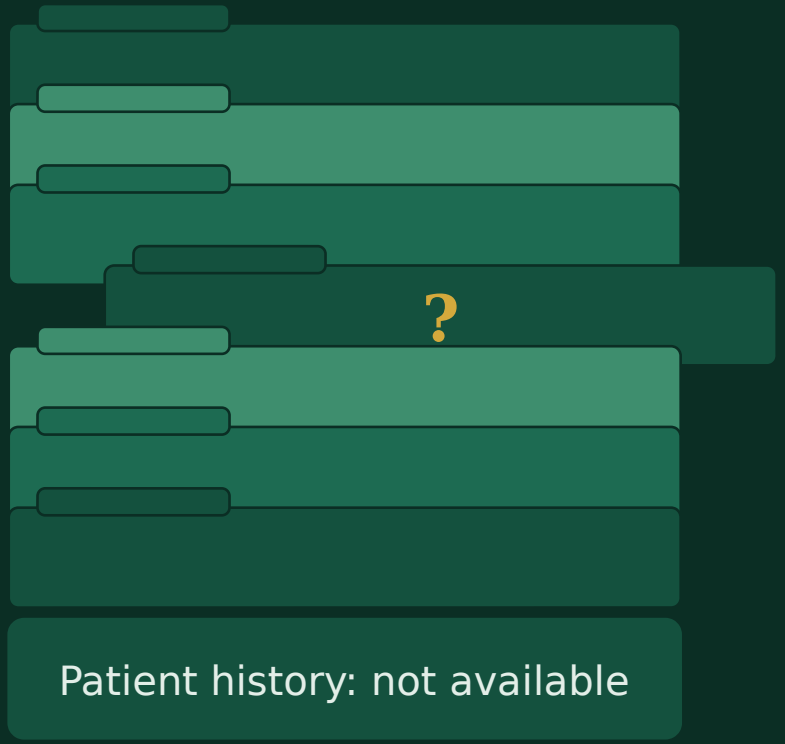


Queue position: **unknown**

The file cannot be found

A new folder is opened and her history starts again from zero.

Earlier diagnoses, allergies and prescriptions are not available to the clinician at the moment they matter most.



Power and network are not guaranteed

- Facilities depend on inverters and generators.
- Internet access drops without warning.
- A system that needs the cloud stops when the connection does.

Design requirement: keep working offline, on the facility LAN



Power infrastructure at a working site

Writing instead of listening

Consultation time is limited, and a large share of it goes to documentation.

What the visit is for

- Listen to the history
- Examine and reason
- Explain the plan to the patient

What the visit becomes

- Handwrite every note
- Re-copy details into records
- Eyes on paper, not on the patient

PART 2 · THE RESPONSE

One connected system

Each problem from the patient's day has a matching module.

1

Registration

Check-in and patient identity

no lost files

2

Queue engine

Ordered tickets, live status

no pushing in

3

EHR

Triage and SOAP notes

history is kept

4

AI scribe

Voice to draft note

less writing

Designed, with event hooks in place, not yet built: Billing · Pharmacy and inventory · Cloud sync

One server per facility

Clients	Reception · Nurse · Doctor · Admin
FastAPI	Core · Registration · Queues · EHR · Scribe
Real-time	Event bus · WebSocket queue updates
Data	PostgreSQL 16 (SQLite in dev) · Alembic

Key decisions

- Runs on the facility LAN, offline-capable
- Modular, so billing and pharmacy plug in
- Docker Compose deploy on one server

The AI scribe runs on-site

Consultation
Doctor and patient speak

›

Whisper
Local speech-to-text

›

Local LLM
Qwen via Ollama drafts a SOAP note

›

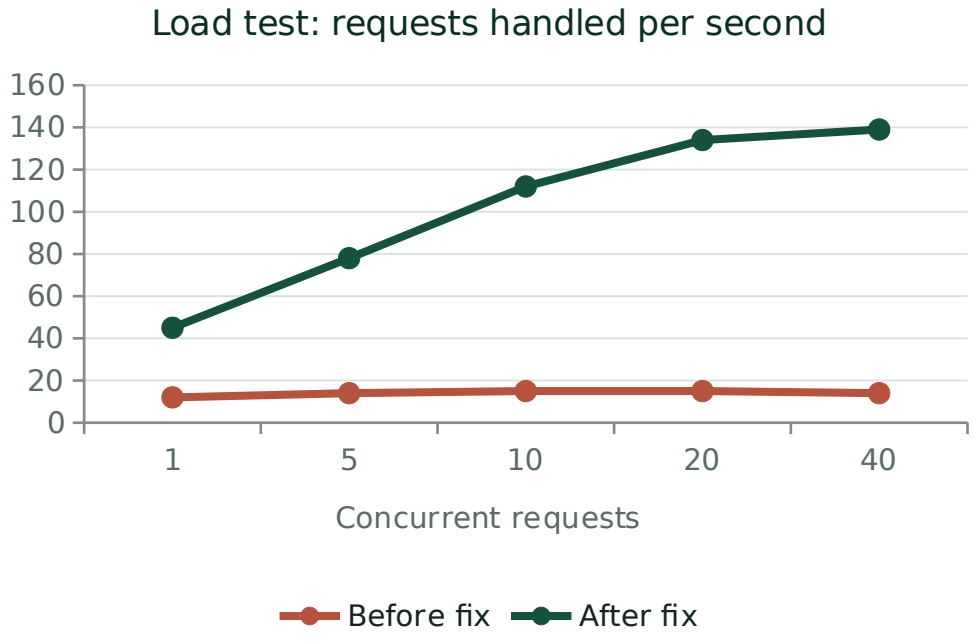
Clinician review
Edit, approve, save to EHR

No cloud API
no per-call cost

Data stays on-site
privacy by design

Clinician in control
draft, never final

Engineered to be trusted



36 automated tests covering flow, queue, scribe and security

Safeguards built in

- Role-based access
- Account lockout after repeated failures
- Refuses weak secrets in production
- Versioned database migrations

CONCLUSION

Her next visit looks different

- ✓ **Registered once** identity on record
- ✓ **Position known** an ordered, visible queue
- ✓ **History on screen** at the point of care
- ✓ **Doctor listens** the scribe drafts the note

Next: billing · pharmacy and inventory · cloud sync



Thank you. Questions welcome.